

Yield-Curve Control Policy under Inelastic Financial Markets

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This paper

A quantitative macroeconomic framework that incorporates

- ① The general equilibrium term-structure of interest rates when financial markets are inelastic with imperfect substitutability
 - Estimation of the cross-elasticity of portfolio rebalancing across maturities
- ② Multiple asset classes (government bonds and private bond) with time-varying, countercyclical term premia
- ③ Endogenous portfolio rebalancing among different kinds of assets
- ④ Government and central bank's explicit balance sheets
- ⑤ A micro-founded welfare criterion with ZLB constraints

which are necessary for quantitative policy experiments (e.g., conventional vs. unconventional monetary policies)

- Focus on yield-curve-control (YCC) policy as a baseline unconventional monetary policy, but the framework can be applied to other unconventional monetary policies as well.

Simple model: the core mechanism

For simplicity, let us assume two bonds only ($f = 1, 2$), rigid prices (no inflation dynamics), no loans to firms, no explicit balance-sheet sizes

Euler equation: the effective savings rate $\hat{r}_{t+1}^S$, not the policy rate $\hat{r}_{t+1}^1$, governs aggregate consumption demand

$$\hat{c}_t = \mathbb{E}_t \left[\hat{c}_{t+1} - \hat{r}_{t+1}^S \right]$$

where

$$\hat{r}_{t+1}^S = \sum_{f=1}^2 \frac{\lambda^f R^f}{R^S} \left(\hat{\lambda}_t^f + \hat{r}_{t+1}^f \right)$$

portfolio return

portfolio share

Finite elasticity of portfolio rebalancing across maturities:

$$\hat{\lambda}_t^1 = \kappa_B \mathbb{E}_t \left[\hat{r}_{t+1}^1 - \hat{r}_{t+1}^2 \right]$$

- Finite κ_B : changes in relative returns move portfolio shares, but not too much
- Conventional policy moves mainly $\hat{r}_{t+1}^1 \implies$ households rebalance along the yield curve

Simple model: conventional policy

Conventional policy: central bank moves the short rate Only

$$\hat{r}_{t+1}^1 = \phi_C \hat{c}_t + \varepsilon_t, \quad \hat{r}_{t+1}^2 = 0$$

- For simplicity, the long-term rate is fixed now ($\hat{r}_{t+1}^2 = 0$). In our quantitative model, there is an incomplete pass-through from the policy rate to the long-term rate.

The effective savings rate under conventional policy is given by:

$$\hat{r}_{t+1}^S = \underbrace{\eta_1}_{\text{Direct pass-through } (\eta_1 < 1)} \hat{r}_{t+1}^1 - \underbrace{\eta_2}_{\text{Composition } (\eta_2 > 0)} \hat{\lambda}_t^1$$

- Tightening shocks $\varepsilon_t > 0$ lead to $\hat{\lambda}_t^1 > 0$, decreasing the effective savings rate $\hat{r}_{t+1}^S$. This effect attenuates the policy-rate channel. $\hat{\lambda}_t^1 > 0$ is determined by the elasticity κ_B .
- Portfolio rebalancing channel of monetary policy

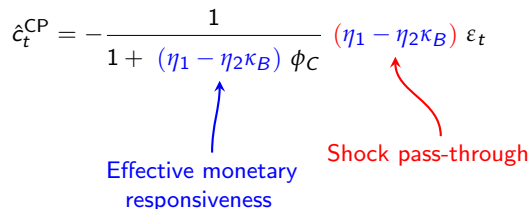
Simple model: conventional policy

Combining the Euler equation, the Taylor rule, and portfolio demand:

$$\hat{c}_t^{\text{CP}} = - \frac{1}{1 + (\eta_1 - \eta_2 \kappa_B) \phi_C} (\eta_1 - \eta_2 \kappa_B) \varepsilon_t$$

Effective monetary responsiveness

Shock pass-through



- Portfolio rebalancing **attenuates** the policy effectiveness as well as the effective policy shock.
- If substitution is strong enough, conventional policy can even have the “wrong” sign

Simple model: why YCC is powerful

Yield-curve-control (YCC) policy: central bank moves both short and long rates following Taylor rules:

$$\hat{r}_{t+1}^1 = \phi_C \hat{c}_t + \varepsilon_t, \quad \mathbb{E}_t \hat{r}_{t+1}^2 = \phi_{C,2} \hat{c}_t$$

Shock pass-through
(Same)



With this, aggregate consumption demand is given by:

$$\hat{c}_t^{\text{YCC}} = - \frac{(\eta_1 - \eta_2 \kappa_B) \varepsilon_t}{1 + \underbrace{(\eta_1 \phi_C + (1 - \eta_1) \phi_{C,2})}_{\text{Policy response}} - \eta_2 \kappa_B \underbrace{(\phi_C - \phi_{C,2})}_{\text{Cross-substitution}}}$$

Two forces (YCC)

- ① **Cross-substitution:** if $\phi_{C,2}$ moves with ϕ_C , households have less incentive to rebalance across maturities
- ② **Policy response:** the entire yield curve reacts to macroeconomic conditions
—YCC targets the exact margin through which conventional policy loses traction

What we do + findings

Takeaway (Conventional vs. YCC)

Yield-curve control (YCC) is powerful in terms of stabilization in both normal and ZLB

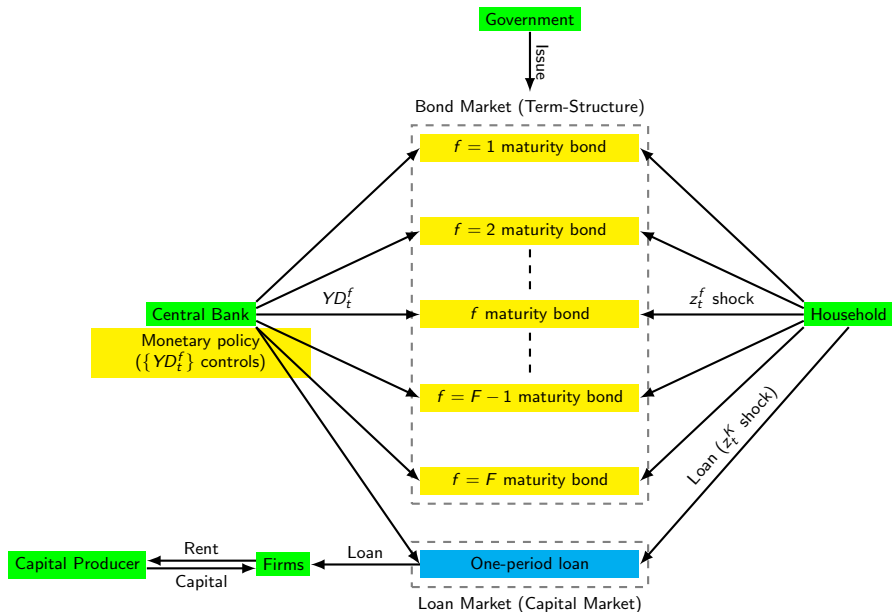
- Other unconventional monetary policies are effective—for example, central bank's direct purchases of private loans is effective^a
- Sometimes, YCC generates a longer ZLB duration

^aThe Fed's Primary and Secondary Market Corporate Credit Facilities in 2020 (Boyarchenko et al., 2022)

Potential mechanism: long term yields↓ $\Rightarrow$ portfolio shares of short bonds and loans↑ $\Rightarrow$ short yields and loan rates↓ (disinflationary for firms) $\Rightarrow$ ZLB duration↑

- Still, YCC is very effective in terms of stabilization, offsetting this channel and reducing the ZLB duration on average.

The model: environment



The model: household

The representative household's problem (given B_0):

subject to

$$C_t + \frac{L_t^H}{P_t} + \frac{\sum_{f=1}^F B_t^{H,f}}{P_t} = \frac{\sum_{f=0}^{F-1} R_t^f B_{t-1}^{H,f+1}}{P_t} + \frac{R_t^K L_{t-1}^H}{P_t} + \int_0^1 \frac{W_t(\nu) N_t(\nu)}{P_t} d\nu + \frac{\Lambda_t}{P_t}$$

$\max_{\{C_{t+j}, N_{t+j}\}}$ **Loans**
 $\mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\log(C_{t+j}) - \left(\frac{\eta}{\eta+1} \right) \left(\frac{N_{t+j}}{\bar{N}_{t+j}} \right)^{1+\frac{1}{\eta}} \right]$
f-maturity rate **Loan rate**
Nominal bond purchase (f-maturity)

where

- ν : intermediate firm index such that:

$$N_t = \left(\int_0^1 N_t(\nu)^{\frac{\eta+1}{\eta}} d\nu \right)^{\frac{\eta}{\eta+1}}$$

- Q_t^f is the nominal price of f -maturity bond with:

$$(\text{Return}) R_t^f = \frac{Q_t^f}{Q_{t-1}^{f+1}}, \quad (\text{Yield}) YD_t^f = \left(\frac{1}{Q_t^f} \right)^{\frac{1}{f}}$$

The model: household and savings

Total savings: $S_t = B_t^H + L_t^H = \sum_{f=1}^F B_t^{H,f} + L_t^H$

Usual bond allocation problem (Ricardian):

$$\max \sum_{f=1}^F \mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^{f-1} B_t^{H,f} \right] \quad \text{s.t.} \quad \sum_{f=1}^F B_t^{H,f} = B_t^H, \quad B_t^{H,f} \geq 0$$

which gives (in equilibrium):

$$\mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^{f-1} \right] = \mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^0 \right], \quad \forall f \implies$$

$$\mathbb{E}_t[\hat{R}_{t+1}^{f-1}] = \hat{R}_{t+1}^0$$

'Expectations hypothesis'

Our approach (Non-Ricardian):

- Split the household into a family $m \in [0, 1]$, each of which decides whether to invest in **bonds** or **loan**, subject to **expectation shock** $\sim$ **Fréchet**
- A **bond** family m is split into members $n \in [0, 1]$, each of whom decides maturity f to invest in, subject to **expectation shock** $\sim$ **Fréchet**

Bond portfolio (e.g., Eaton and Kortum (2002))

$$\lambda_t^{HB,f} \equiv \left(\frac{z_t^f \mathbb{E}_t [Q_{t,t+1} R_{t+1}^{f-1}]}{\Phi_t^B} \right)^{\kappa_B}$$

κ_B Portfolio preference shock

where f -maturity share

$$\Phi_t^B \equiv \left[\sum_{j=1}^F \left(z_t^j \mathbb{E}_t [Q_{t,t+1} R_{t+1}^{j-1}] \right)^{\kappa_B} \right]^{\frac{1}{\kappa_B}}$$

- Deviate from expectation hypothesis $\implies \exists$ downward-sloping demand curve after log-linearization with finite demand elasticity and cross-elasticities
- Shape parameter κ_B : (inverse of) a degree of bonds market segmentation
- $z_t^f = 1$, $\kappa_B \rightarrow \infty$, then again expectations hypothesis (i.e., Ricardian)

Effective bond market rates

$$R_{t+1}^{HB} = \sum_{f=0}^{F-1} \lambda_t^{HB,f+1} R_{t+1}^f$$

Loan share (e.g., Eaton and Kortum (2002))

Portfolio preference shock

$$\lambda_t^K = \left(\frac{z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K]}{\Phi_t^S} \right)^{\kappa_S}$$

Loan share

where

$$\Phi_t^S = \left[\left(\mathbb{E}_t [Q_{t,t+1} R_{t+1}^{HB}] \right)^{\kappa_S} + \left(z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K] \right)^{\kappa_S} \right]^{\frac{1}{\kappa_S}}$$

- $\exists$ downward-sloping demand curve after log-linearization (for bonds and private loan) with finite demand elasticity and cross-elasticities
- Shape parameter κ_S : (inverse of) a degree of market segmentation between government bonds vs loan

Effective savings rate: governs intertemporal substitution

$$\begin{aligned} R_t^S &= (1 - \lambda_{t-1}^K) R_t^{HB} + \lambda_{t-1}^K R_t^K \\ &= (1 - \lambda_{t-1}^K) \sum_{f=0}^{F-1} \lambda_{t-1}^{HB, f+1} R_t^f + \lambda_{t-1}^K R_t^K \end{aligned}$$

Enters Euler equation

Equilibrium + market clearing

Bond market equilibrium:

$$B_t^{H,f} + B_t^{G,f} + B_t^{CB,f} = 0, \quad \forall f = 1, \dots, F$$

► Capital Producer, Firms, and Government

Monetary
policy

Central bank: monetary policy through balance sheet adjustments

- **Conventional:** Taylor rules on YD_t^1 (only adjusting $B_t^{CB,1}$)—still, $B_t^{CB,f}$ for $f > 1$ are subject to a quantity shock $u_t^{CB,f}$
- **Yield-curve-control (YCC):** Taylor rules on $\{YD_t^f\}$ (adjusting $\{B_t^{CB,f}\}$)
- The central bank holds corporate loans L_t^{CB} , which is subject to a quantity shock u_t^L
- Subject to **zero lower bound (ZLB)**

► Conventional ► Unconventional

Steady-state U.S. calibrated yield curve (up to 30 years)

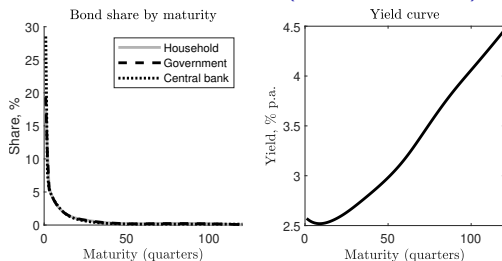


Figure: Steady-state bond portfolios of households, government, and central bank and the resultant yield curve (December 2002 - June 2007)

Estimation: $\kappa_B = 10$ from the aggregate bond portfolio data

» Estimation

Calibration: given $\kappa_B = 10$ and $\kappa_S = 6$:

- $\{z^f\}_{f=1}^F$ (i.e., maturity preference for a maturity- f) $\Rightarrow$ yield curve slopes
- z^K (i.e., preference for private loan) $\Rightarrow$ the yield curve level
- Result: $z^1 = 1 \gg z^f$ for $f \geq 2$ (e.g., **safety - liquidity** premium)

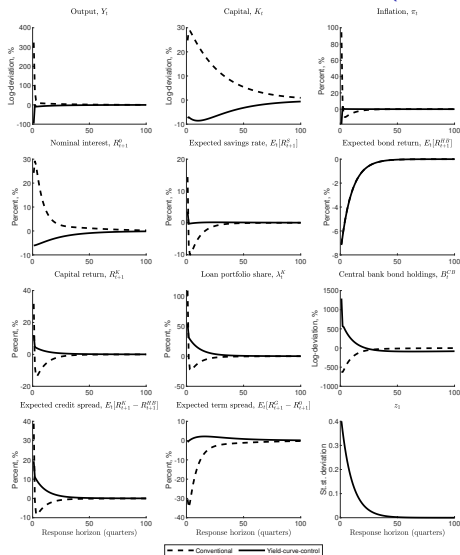
» Supply effects

» Demand effects

» Deficit ratio

Short-run analysis (Impulse-responses)

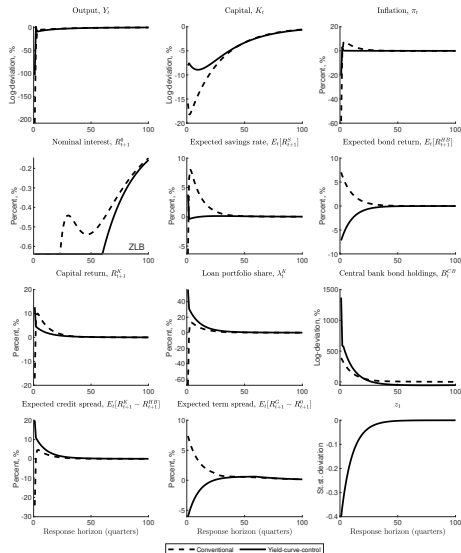
A shock to the preference for the short-term bond (impulse response to z_t^1)



Countercyclical term premia

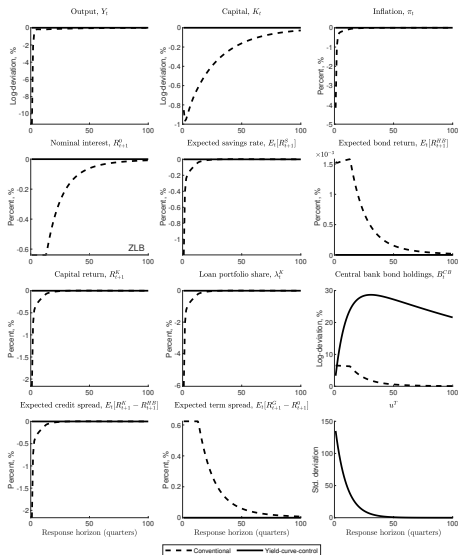
- YCC is almost perfectly stabilizing (filling gaps in bond demand)

ZLB impulse response to z_t^1



YCC still is powerful, but duration of ZLB episodes $\uparrow$ in this case

ZLB impulse response to an exogenous tax hike



With **conventional** policy: non-Ricardian (tax $\uparrow \Rightarrow$ bond supply $\downarrow$)

Policy comparison (Conventional, Yield-Curve-Control, and Mixed)

We also consider:

- **Mixed policy**: central bank starts controlling long-term rates only when FFR hits ZLB, thus **YCC** only at the ZLB

	Conventional	Yield-Curve-Control	Mixed Policy
Mean ZLB duration	4.425 quarters	2.375 quarters	2.9402 quarters
Median ZLB duration	2 quarters	2 quarters	1 quarters
ZLB frequency	14.303%	0.19192%	8.9394%
Welfare	-1.3425%	-0.99039%	-1.4067%

Table: Policy comparisons (ex-ante)

ZLB duration: YCC < Mixed < Conventional

ZLB frequency: YCC < Mixed < Conventional

Welfare: Mixed < Conventional < YCC

Thank you very much!
(Appendix)

Key previous works (only a few among many) [▶▶ Go back](#)

- The term-structure and macroeconomy: [Ang and Piazzesi \(2003\)](#), [Rudebusch and Wu \(2008\)](#), [Bekaert et al. \(2010\)](#)
- Central bank's endogenous balance sheet size as an another form of monetary policy: [Gertler and Karadi \(2011\)](#), [Cúrdia and Woodford \(2011\)](#), [Christensen and Krogstrup \(2018, 2019\)](#), [Karadi and Nakov \(2021\)](#), [Sims and Wu \(2021\)](#)
- Time-varying term premia in a New Keynesian model without inelastic bond markets: [Rudebusch and Swanson \(2012\)](#), [Buncic and Lentner \(2016\)](#), [Kliem and Meyer-Gohde \(2022\)](#), [Mertens and Zhang \(2025\)](#), [Gurkaynak and Wright \(2012\)](#)
- ZLB and the supply of safe bonds: [Swanson and Williams \(2014\)](#), [Caballero and Farhi \(2017\)](#), [Caballero et al. \(2021\)](#), [Li and Merkel \(2025\)](#),
- Preferred-habitat term-structure (and limited risk-bearing): [Greenwood et al. \(2020\)](#), [Vayanos and Vila \(2021\)](#), [Ray \(2019\)](#), [Droste, Gorodnichenko, and Ray \(2021\)](#), [Gourinchas et al. \(2021\)](#), [Kekre et al. \(2023\)](#)
- Cross-elasticity and portfolio rebalancing: [Alpanda and Kabaca \(2020\)](#), [Jansen et al. \(2025\)](#)

Our paper: General equilibrium term-structure under inelastic financial markets + balance sheet quantities of the government and central bank + yield-curve control (YCC) + novel ways to generate and estimate market segmentation

Bond family m : a member n has the following expectation shock:

$$\mathbb{E}_{m,n,t} \left[Q_{t,t+1} R_{t+1}^{f-1} \right] = z_{n,t}^f \cdot \mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^{f-1} \right], \quad \forall f = 1, \dots, F$$

with $z_{n,t}^f$ follows a **Fréchet** distribution with location parameter 0, scale parameter z_t^f , and shape parameter κ_B

Aggregation (**Eaton and Kortum (2002)**)

$$\begin{aligned} \lambda_t^{HB,f} &\equiv \mathbb{P} \left(\mathbb{E}_{m,n,t} \left[Q_{t,t+1} R_{t+1}^{f-1} \right] = \max_j \left\{ \mathbb{E}_{m,n,t} \left[Q_{t,t+1} R_{t+1}^{j-1} \right] \right\} \right) \\ &= \left(\frac{z_t^f \mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^{f-1} \right]}{\Phi_t^B} \right)^{\kappa_B} \end{aligned}$$

f -maturity share

- Deviate from expectation hypothesis $\implies \exists$ downward-sloping demand curve after log-linearization with finite demand elasticity and cross-elasticities
- Shape parameter κ_B : (inverse of) a degree of bonds market segmentation

Effective bond market rates

$$R_{t+1}^{HB} = \sum_{f=0}^{F-1} \lambda_t^{HB,f+1} R_{t+1}^f$$

Loan vs. bond decision: a family m solves the following problem

$$\max \mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^{HB} B_{m,t}^H \right] + z_{m,t}^K \cdot \mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^K L_{m,t} \right] \quad \text{s.t.}$$

$$B_{m,t}^H + L_{m,t} = S_t, \quad B_{m,t}^H \geq 0, \quad \text{and} \quad L_{m,t} \geq 0$$

with $z_{m,t}^K$ follows a Fréchet distribution with location parameter 0, scale parameter z_t^K , and shape parameter κ_S

Aggregation (Eaton and Kortum (2002))

$$\lambda_t^K = \left(\frac{z_t^K \mathbb{E}_t [Q_{t,t+1} R_{t+1}^K]}{\Phi_t^S} \right)^{\kappa_S}$$

Loan share



- $\exists$ downward-sloping demand curve after log-linearization (for bonds and private loan) with finite demand elasticity and cross-elasticities
- Shape parameter κ_S : (inverse of) a degree of market segmentation between government bonds vs loan

Effective savings rate: governs intertemporal substitution

$$R_t^S = \left(1 - \lambda_{t-1}^K \right) R_t^{HB} + \lambda_{t-1}^K R_t^K$$

$$= \left(1 - \lambda_{t-1}^K \right) \sum_{f=0}^{F-1} \lambda_{t-1}^{HB,f+1} R_t^f + \lambda_{t-1}^K R_t^K$$

Capital producer: lend capital to intermediate firms at price P_t^K (competitive)

Firms: standard with Cobb-Douglas production (pricing à la **Calvo** (1983))

- **One financial friction:** firms secure private loans from the household and the central bank to operate: for simplicity, borrow γ portion of the revenue it generates:

$$\underbrace{L_t(\nu)}_{\text{Loan of firm } \nu} \geq \gamma(1 + \zeta^F) P_t(\nu) Y_t(\nu), \forall \nu$$

Government: with the following budget constraint

$$\frac{B_t^G}{P_t} = \frac{R_t^G B_{t-1}^G}{P_t} - \left[\underbrace{\zeta_t^G}_{\substack{\uparrow \\ \frac{G_t}{Y_t} \text{ (Exogenous)}}} + \underbrace{\zeta^F}_{\text{Production subsidy}} - \underbrace{\zeta_t^T}_{\substack{\uparrow \\ \frac{T_t}{Y_t} \text{ (Exogenous)}}} \right] Y_t, \quad R_t^G = \sum_{f=0}^{F-1} \underbrace{\lambda_{t-1}^{G,f+1}}_{\substack{\uparrow \\ \text{(Exogenous)}}} R_t^f$$

- **Government:** a natural issuer of the entire bond market

Conventional monetary policy

Under the **conventional** monetary policy, central banks set Taylor rules on YD_t^1 while not manipulating longer term bonds holdings

- Long-term yields fluctuate endogenously (in response to shocks + changes in short-term rate)

$$R_{t+1}^0 \equiv YD_t^1 = \max \left\{ YD_t^{1*}, \underset{\text{ZLB}}{1} \right\}$$

$$YD_t^{1*} = \overline{YD}^1 \left(\frac{YD_{t-1}^{1*}}{\overline{YD}^1} \right)^{\rho_1} \left(\frac{YD_{t-2}^{1*}}{\overline{YD}^1} \right)^{\rho_2} \underbrace{\left(\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\gamma_\pi^1} \left(\frac{Y_t}{\bar{Y}} \right)^{\gamma_y^1} \right)}_{\text{Targeting}} \cdot \exp \left(\underset{\text{MP shock } (f=1)}{\tilde{\varepsilon}_t^{YD^1}} \right)^{1-(\rho_1+\rho_2)}$$

$$\underbrace{\frac{B_t^{CB,f}}{A_t \bar{N}_t P_t}}_{\text{Normalized holding of } f > 1 \text{ fixed}} = \frac{\overline{B^{CB,f}}}{\overline{A \bar{N} P}} \exp \left(\underset{\text{Quantity shock}}{\tilde{u}_t^{CB,f}} \right) \quad \forall f = 2, \dots, F$$

Unconventional monetary policy: yield-curve-control (YCC)

In the **unconventional** monetary policy case, central bank targets all yields along the yield curve, assuming the Taylor-type rule for each maturity yield

- Back out the needed purchases of each maturity $\forall f$, which are endogenous

$$R_{t+1}^0 \equiv YD_t^1 = \max \left\{ YD_t^{1*}, \underset{\text{ZLB}}{1} \right\}$$

$$YD_t^{1*} = \overline{YD}^1 \left(\frac{YD_{t-1}^{1*}}{\overline{YD}^1} \right)^{\rho_1} \left(\frac{YD_{t-2}^{1*}}{\overline{YD}^1} \right)^{\rho_2} \left(\underbrace{\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\gamma_\pi} \left(\frac{Y_t}{\bar{Y}} \right)^{\gamma_y}}_{\text{Targeting}} \cdot \exp \left(\tilde{\varepsilon}_t^{YD^1} \right) \right)^{1-(\rho_1+\rho_2)}$$

MP shock ($f = 1$)

$$YD_t^{f*} = \overline{YD}^f \left(\frac{YD_{t-1}^{f*}}{\overline{YD}^f} \right)^{\rho_1} \left(\frac{YD_{t-2}^{f*}}{\overline{YD}^f} \right)^{\rho_2} \left(\underbrace{\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\gamma_\pi} \left(\frac{Y_t}{\bar{Y}} \right)^{\gamma_y}}_{\text{Targeting}} \cdot \exp \left(\tilde{\varepsilon}_t^{YD^f} \right) \right)^{1-(\rho_1+\rho_2)}$$

MP shock ($\forall f \geq 2$)

Central bank's corporate loan policy

The central bank sets its corporate loan holdings L_t^{CB} —the loan rate R_t^K adjusts endogenously to clear the loan market:

$$\frac{L_t^{CB}}{A_t \bar{N}_t P_t} = \overline{\left(\frac{L^{CB}}{A \bar{N} P} \right)} \cdot \exp \left(u_t^L \right),$$

Quantity shock
(corporate loans)

» Go back

Estimation of κ_B

From portfolio equations:

$$\lambda_t^{HB,f} = \left(\frac{z_t^f \mathbb{E}_t \left[Q_{t,t+1} R_{t+1}^{f-1} \right]}{\Phi_t^B} \right)^{\kappa_B}$$

leading to:

f-maturity share

$$\log \left(\lambda_t^{H,f} \right) - \log \left(\lambda_t^{H,l} \right) = \alpha^f + \kappa_B \cdot \mathbb{E}_t \left[r_{t+1}^{f-1} - r_{t+1}^{l-1} \right] + \varepsilon_t^f \quad (1)$$

Local projection à la **Jordà (2005)**:

$$\log \left(\lambda_{t+h}^{H,f} \right) - \log \left(\lambda_{t+h}^{H,l} \right) = \alpha_h^f + \kappa_{B,h} \cdot \left[yd_t^f - yd_t^l \right] + \mathbf{x}_t' \beta_h^f + \varepsilon_{t+h}^f, \quad h \geq 0$$

- Long maturity: $f = 5 \sim 10$ years and short: $l = 15 \sim 90$ days (bunching) for portfolio shares and use $f = 7$ years and $l = 3$ months for yields
- Control variables (e.g., lagged $\log \left(\lambda_{t-1}^{H,f} \right) - \log \left(\lambda_{t-1}^{H,l} \right)$ for serial correlation)

Local projection à la **Jordà (2005)**:

$$\log \left(\lambda_{t+h}^{H,f} \right) - \log \left(\lambda_{t+h}^{H,l} \right) = \alpha_h^{fl} + \kappa_{B,h} \cdot \left[yd_t^f - yd_t^l \right] + \mathbf{x}_t' \beta_h^{fl} + \varepsilon_{t+h}^{fl}, \quad h \geq 0$$

Instrument for the yield spread $yd_t^f - yd_t^l$:

- Lagged yield spread $yd_{t-1}^f - yd_{t-1}^l$ ($\perp$ demand shocks, e.g., z_t^f, z_t^l)
- Instrument with the Treasury supply shocks of **Phillot (2025)**, which are identified from high-frequency movements in Treasury futures prices around auction announcements.

►► Go back

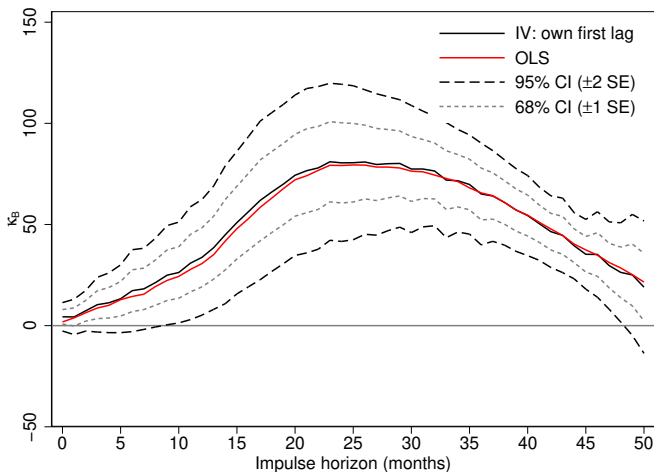


Figure: Impulse-Response to the yield spread $yd_t^f - yd_t^l$, instrumented with the lagged yield spread $yd_{t-1}^f - yd_{t-1}^l$. The sample period is from 2003m3 to 2019m3.

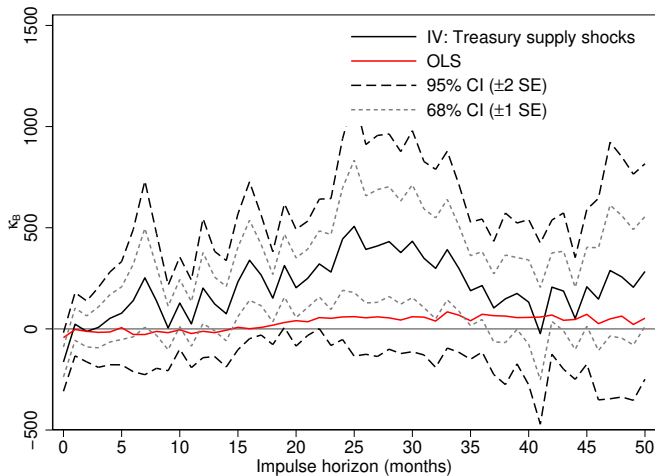


Figure: Impulse-Response to the yield spread $yd_t^f - yd_t^l$, instrumented with **Phillot (2025)**'s 5- and 10-year Treasury supply shocks. The sample period is from 2003m3 to 2019m3.

Government's bond supply effects

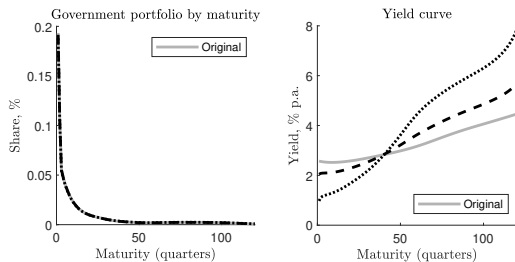


Figure: Government's bond issuance portfolio and yield curve

- Government's supply of f -maturity bond $\uparrow \Rightarrow$ its yield $\uparrow$ (i.e., price effect)
- Similar to Krishnamurthy and Vissing-Jorgensen (2011) and D'Amico and King (2013) in the long run

Central bank's bond demand effects

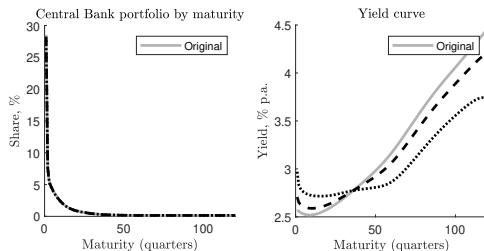


Figure: Central bank's bond demand portfolio and yield curve

- **Inelastic** markets $\Rightarrow$ central bank's bond demand matters in the long run

A deficit ratio: comparative statics

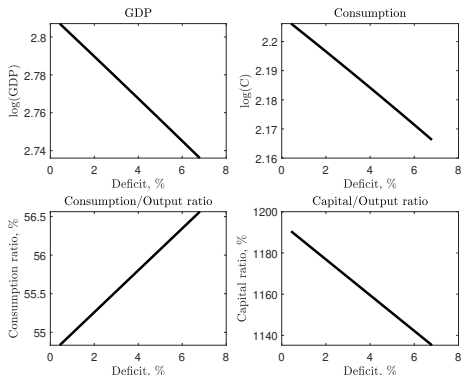


Figure: Variations in a deficit ratio $\zeta_t^G + \zeta_t^F - \zeta_t^T$

A higher deficit ratio $\Rightarrow$ depressed economy (for $R^G \downarrow$)

Go back

A deficit ratio: comparative statics

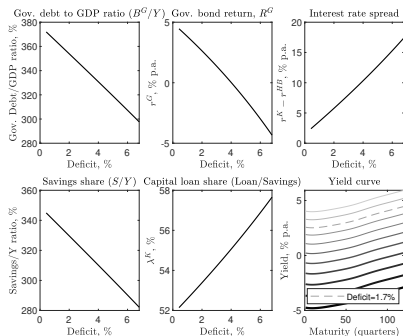
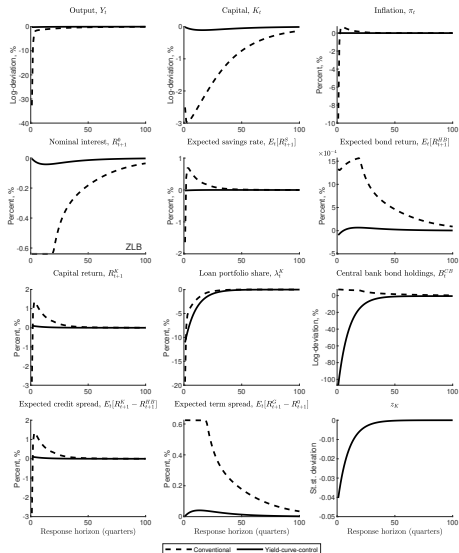


Figure: Variations in a deficit ratio $\zeta_t^G + \zeta^F - \zeta_t^T$

A higher deficit ratio $\Rightarrow$ depressed economy (for $R^G \downarrow$)

- An entire yield curve $\downarrow$

ZLB impulse response to z_t^K



Impulse-response to an exogenous tax hike shock

